

Orvin Andika Ikhsan Abhista

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Professional Experience

STEI ITB - Algorithm Strategy Course Assistant - Designed and reviewed 5+ major assignments for algorithm courses, improving question clarity and alignment with learning objectives. - Collaborated with lecturers to refine problem difficulty and ensure coverage of key topics such as dynamic programming and graph algorithms.	Feb 2026 – Present
STEI ITB - Discrete Mathematics Course Assistant - Developed and validated 5+ quizzes, ensuring logical correctness and conceptual depth. - Assisted in grading and provided detailed feedback to 100+ students to enhance understanding.	Feb 2026 – Present
STEI ITB- Linear and Geometric Algebra Course Assistant Created and reviewed 5+ assignments and quizzes covering linear and geometric algebra..	Sep 2025 – Dec 2025
PTPB ITB - Computational Thinking Course Assistant - Designed and evaluated project-based assignments focusing on problem decomposition and algorithmic thinking. - Guided 100+ students in developing structured solutions using pseudocode and basic programming.	Sep 2025 – Dec 2025
STEI ITB - Computer Science and Engineering Laboratory Assistant - Coordinated academic support across multiple core courses under the laboratory. - Supported course operations by aligning assignment standards, timelines, and evaluation approaches across courses. - Contributed to improving academic processes and resource organization within the laboratory.	Sep 2025 – Present

Education

Institut Teknologi Bandung - B. Eng in Informatics CGPA: 3.88/4.00	Aug 2023 - Present
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Organizational Experience

DPP HMIF ITB Vice Chairman of the Regeneration Commision - Led regeneration initiatives to prepare the next leadership of DPP HMIF, ensuring continuity and sustainability of the organization. - Organized and executed HMIF general election end-to-end, managing candidate preparation, voting processes, and result validation. - Coordinated with multiple stakeholders to ensure a transparent, fair, and well-governed election process.	Mar 2025 – Mar 2026
DPP HMIF ITB Staff of the System Improvement Commision Conducted organizational system evaluations through research and benchmarking studies with other student organizations.	Oct 2024 – Mar 2025

Projects & Achievements

Nimonspedia - E-Commerce App - Developed a full-stack e-commerce application with product catalog, cart, and transaction management features. - Implemented user authentication and optimized data flow to ensure smooth shopping experience.	
IRK Library Android App for IRK Library Course Material - Built an Android application to centralize IRK course materials and improve student access to learning resources. - Implemented built-in calculators (e.g., matrix operations calculator) to support problem solving directly within the app. - Enhanced usability through structured navigation and integrated tools for both theory and computation	
Little Alchemy 2 Recipe Finder - Developed an algorithm-based tool to efficiently find crafting combinations in Little Alchemy 2. - Optimized search performance using graph traversal techniques to reduce lookup time.	
O.W.C.A. Logistics - Designed a logistics management system to handle delivery tracking and resource allocation. - Improved operational visibility through structured data handling and workflow organization.	

ATS CV Finder

- Built a tool to analyze and match CVs against job descriptions using keyword extraction and scoring methods.
- Improved candidate filtering efficiency by automating resume screening processes.

Raft Consensus Algorithm Implementation

- Implemented the Raft distributed consensus algorithm to ensure fault-tolerant and consistent state replication.
- Simulated leader election, log replication, and failure handling in a distributed system environment.

AI-Based Course Scheduling Optimizer

- Developed an AI-driven scheduler to optimize course timetables under multiple constraints.
- Applied optimization/search algorithms to minimize conflicts and improve scheduling efficiency.

ChatTCP - TCP-over-UDP

- Built a reliable communication protocol over UDP by implementing TCP-like features (3-way handshake, retransmission, sequencing).
- Ensured reliable data delivery and connection management in a custom networking system.

Quadtree Compressor

- Implemented image compression using quadtree data structures to reduce storage while preserving quality.
- Optimized recursive partitioning for efficient spatial representation.

Judol Detector

- Developed a text classification system to detect and filter online gambling-related comments on YouTube.
- Improved moderation efficiency by automatically flagging high-risk comments for review.

Chess End Game Solver

- Developed a chess endgame solver using minimax with alpha-beta pruning.
- Improved decision efficiency and accuracy in evaluating optimal moves for constrained game states.

Skills

Programming Languages:	Python, Java, Javascript, PHP,Go, Typescript, SQL, C, C++, Kotlin.
Technologies:	React.js, Next.js, Node.js, MySQL, Postgresql, Github, Postman, Figma
Skills:	Nginx, gRPC, Redis, Docker
Languages:	English (proficient), Indonesian (native)